# AMES HOUSING CAPSTONE PROJECT

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## INTRODUCTION

In this project, I utilized the Ames Housing dataset, which contains detailed information regarding residential properties in Ames, Iowa. The dataset serves as a benchmark for house price prediction, featuring over 80 attributes including house size, quality, location, and various structural details.

**The primary objectives of this analysis were:**

- To identify the most significant factors influencing property sale prices.
- To apply a complete data pipeline, including data cleaning, feature engineering, and analysis.
- To uncover patterns that explain differences in house prices.

## CLEANING SUMMARY

**Before performing any analysis, the dataset required several cleaning steps to ensure data quality and reliability.**

- **Handling Missing Values:** Missing values were handled based on the data type. For numerical columns such as Lot Frontage, I used the median because it is less affected by outliers. For categorical columns like Alley, I replaced missing values with "Unknown", as missing values often indicate that the feature does not exist.

- **Data Type Correction:** I converted columns such as MS SubClass from numerical to string format because they represent categories rather than numerical values.

- **Outlier Management:** I used a boxplot to inspect SalePrice and capped extreme values at the 99th percentile. This helps reduce the effect of extreme values while keeping the dataset realistic.

- **Integrity Checks:** I confirmed that there were no duplicate rows and added checks to ensure SalePrice contains no missing values and all values are positive.

## FEATURE ENGINEERING SUMMARY

**After cleaning the data, I created new features and transformations to improve the dataset.**

- **New Features:** I created price_per_sqft by dividing SalePrice by the living area to allow fair comparison between houses of different sizes. I also created bathroom_ratio to represent the relationship between bathrooms and total rooms.

- **Interaction Feature:** I created quality_x_area by multiplying Overall Quality and Living Area. This captures how size and quality together affect house prices.

- **Encoding:** I applied one-hot encoding to categorical variables such as Neighborhood and MS Zoning. I also used ordinal encoding for Kitchen Quality to preserve its order.

- **Scaling and Transformation:** I applied a log transformation to SalePrice to reduce skewness and make the distribution more balanced. I also scaled numerical features using StandardScaler to ensure consistent ranges across features.

## KEY FINDINGS

**Based on the correlation heatmap and visual analysis, several important patterns were identified:**

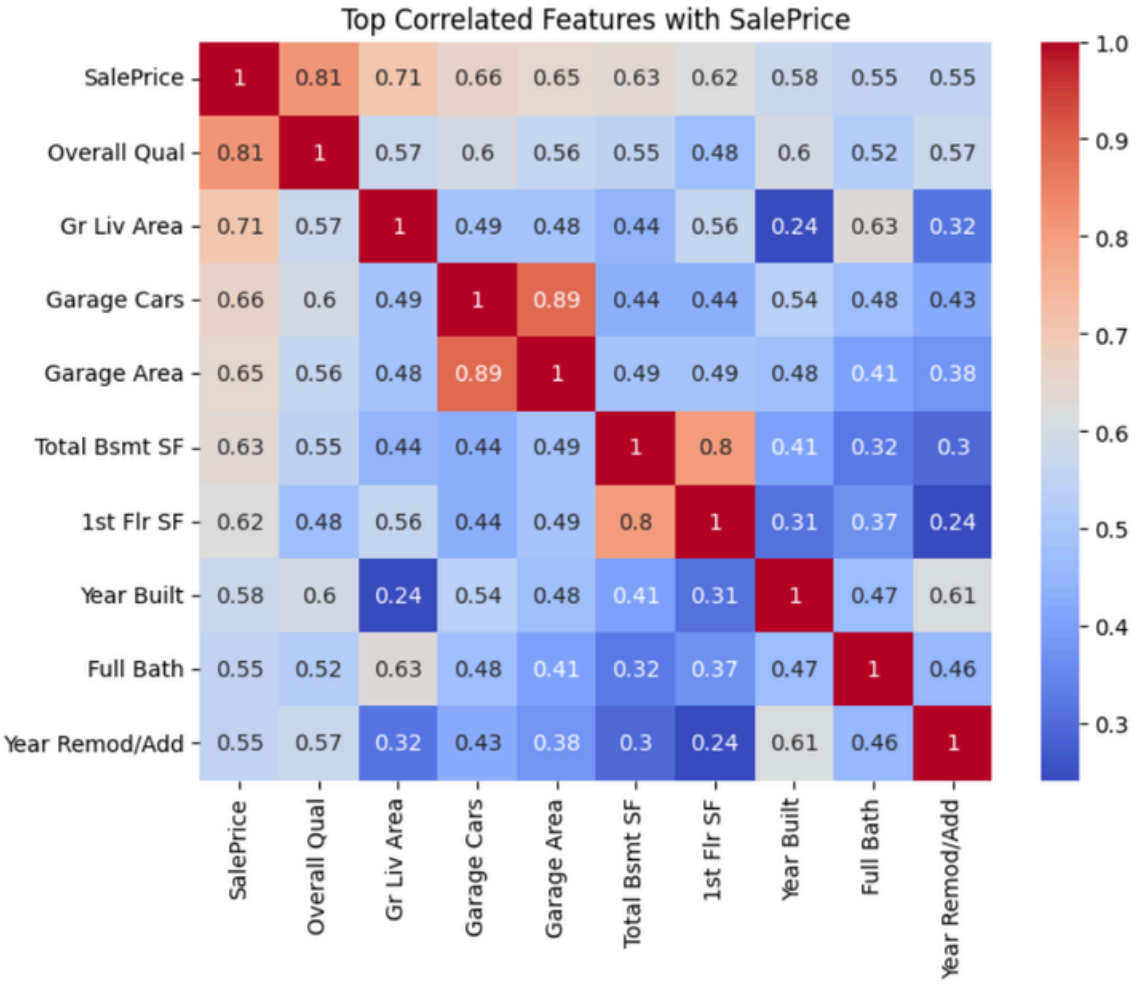
**1. Quality is the Dominant Factor:** Overall Quality has the strongest positive relationship with SalePrice. Higher quality houses are significantly more expensive.

**2. Living Area is Important:** There is a strong positive relationship between living area and price. Larger houses generally have higher prices, especially when combined with high quality.

**3. Newer Houses Tend to Cost More:** Newer houses generally have higher prices compared to older ones.

**4. Neighborhood Impact:** Different neighborhoods show noticeable differences in average house prices, indicating that location plays an important role.

## CORRELATION HEATMAP




## WHAT I WOULD DO NEXT

**If I had more time, I would extend this project in several ways:**

- Build machine learning models such as Linear Regression or Random Forest to predict house prices.
- Apply cross-validation and improve feature selection to increase model accuracy.
- Explore additional data sources, such as location-based data, to better understand neighborhood effects.